

BEE 208 — Question 12: Solar Panel Voltage Tracker

Control Statement: while loop | Faculty of Engineering, ATU

1. ALGORITHM

Step 1: Start

Step 2: Declare variable voltage (double)

Step 3: Prompt user to enter first voltage reading

Step 4: Read voltage

Step 5: While voltage ≥ 18 , display 'Voltage is within operating level'

Step 6: Prompt and read next voltage reading

Step 7: When voltage < 18 , exit loop

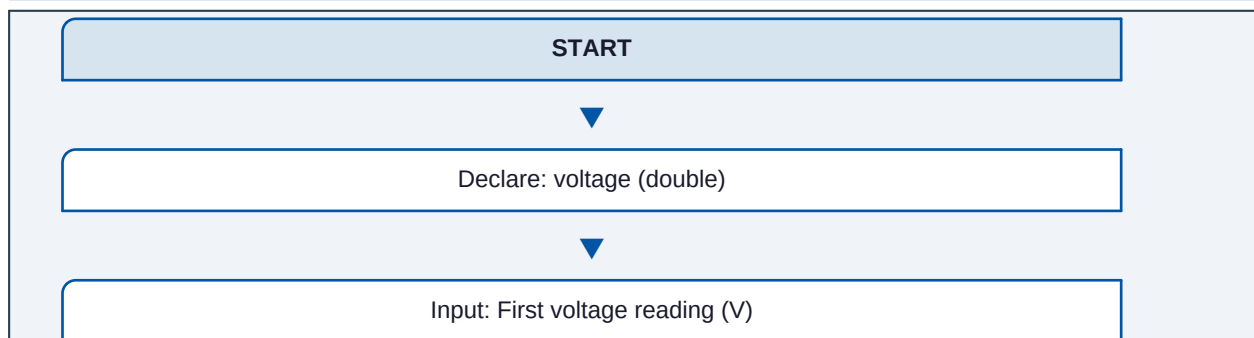
Step 8: Display 'Solar panel voltage below operating level.'

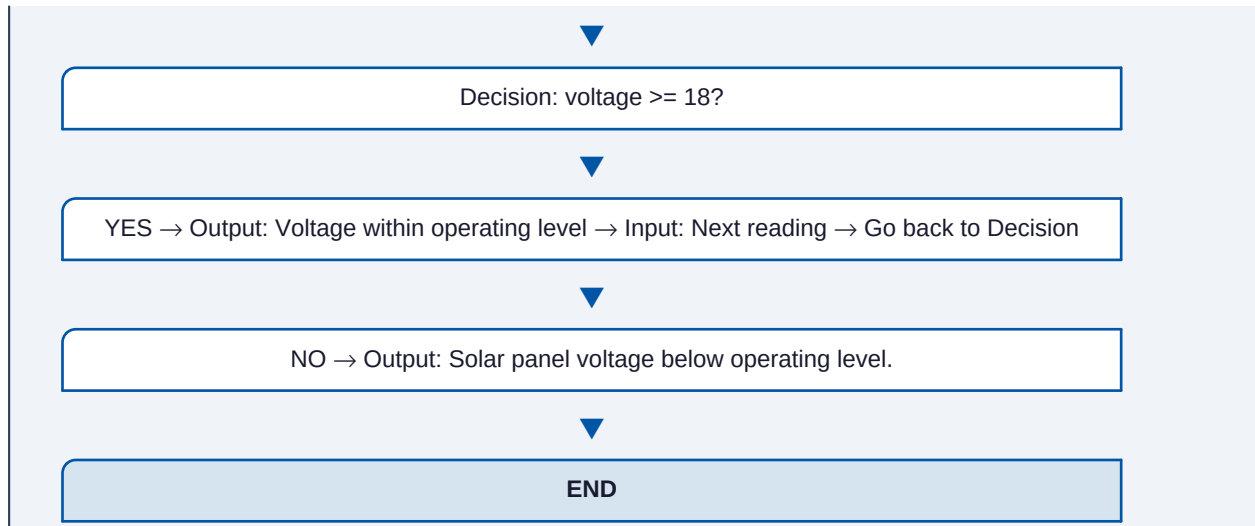
Step 9: End

2. PSEUDOCODE

```
BEGIN  
  
DECLARE voltage AS DOUBLE  
  
OUTPUT 'Enter first voltage reading (V):'  
  
INPUT voltage  
  
WHILE voltage  $\geq 18$  DO  
  
    OUTPUT 'Voltage within operating level:', voltage, 'V'  
  
    OUTPUT 'Enter next voltage reading (V):'  
  
    INPUT voltage  
  
END WHILE  
  
OUTPUT 'Solar panel voltage below operating level.'  
  
END
```

3. FLOWCHART (Structured Diagram)





4. TEST RESULTS

Test Item	Value / Description	Test Item	Value / Description
Input	22, 20, 19, 17	Expected Output	OK×3, then: Voltage below operating level.
Input	15 (first entry)	Expected Output	Immediately: Voltage below operating level.